

Analyzing Urban Tree Planting Across Boston

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Project Summary

Over the semester, I had the privilege of working with TreeBoston. I volunteered to plant trees in an underrepresented neighborhood of the city and crafted the next couple of visualizations I'll present. TreeBoston, a nonprofit organization based in Boston, aims to plant over 100,000 trees across the city. Their goal is not only to improve air quality but also to prioritize underserved neighborhoods like Roxbury and Dorchester. Each tree planted helps reduce air pollution, lowers urban temperatures, and decreases storm water runoff, making TreeBoston's work incredibly important.

As mentioned earlier, I worked on a couple of visualizations over the past month. I used two primary datasets: the Boston Open Spaced shapefile and the BPRD Trees dataset, primarily joined on tree type. I ended up with 521 records that included species

names, updated geographic coordinates, and the year the open space was acquired for the tree. Since the date_plant field was largely unusable, I substituted YearAcquired for open spaces as a temporal placeholder. After filtering and creating my visualizations using Tableau, I discovered that red maple is the most planted species, followed by Pin Oak and flowering dogwood. I also noticed that a significant portion of the trees planted had more concentrated populations in Roxbury, Mattapan, and Dorchester which TreeBoston is actively working to increase.

Although creating visualizations was enjoyable, volunteering at New Calvary Cemetery in Mattapan was an even more thrilling experience. I planted sawtooth oak trees and connected with the workers and community members who shared my passion for making Boston greener. Overall, it was an amazing experience, and I would do it again in a heartbeat.

Data

Boston Open Spaces Dataset, which provided geographic boundaries for each tree, served as my initial source. Additionally, the BRPD Trees dataset, which included individual tree records with their species location and planting information, formed the backbone of my analysis. I merged these datasets in Tableau, joining them on the OS_ID column. I also utilized a Boston Neighborhoods Shapefile to outline the various neighborhoods in Boston.

The original datasets had a few issues that required cleaning. One issue was the date_plant column in the BPRD Trees dataset, which was nearly empty with only two values. As a workaround, I used the YearAcquired column from the Open Spaces dataset as a placeholder for that column. Another issue was the Species column, which contained numerous data points with "Empty Pit/Planting Site." I had to filter out these entries because they caused abnormalities in my Tableau visualizations. Longitude and

latitude had to be corrected in my graphs because the coordinates were not precise in the initial rendition. I had to correct them to the updated ones provided by my classmates.

While Tree Boston is one of my favorite organizations, their data storage and collection processes could be improved. One recommendation that would make future data analyst visualizations much clearer is to make the date_plant column a required field with a standardized format, such as DD-MM-YYYY. Another recommendation is to validate the coordinates of entries to avoid map erosion, as we did during the semester. The data types I had to work with were nominal/categorical, which included species names, neighborhood names, and open space names. In addition, the count of trees per species and the count of open spaces fell under quantitative discrete data, while latitude, longitude, and year acquired were under qualitative continuous data. I had multiple geographic data, including coordinates for the map plots. Lastly, I had a workaround for the date field, creating a pseudo-temporal data column called YearAcquired.

Task Analysis

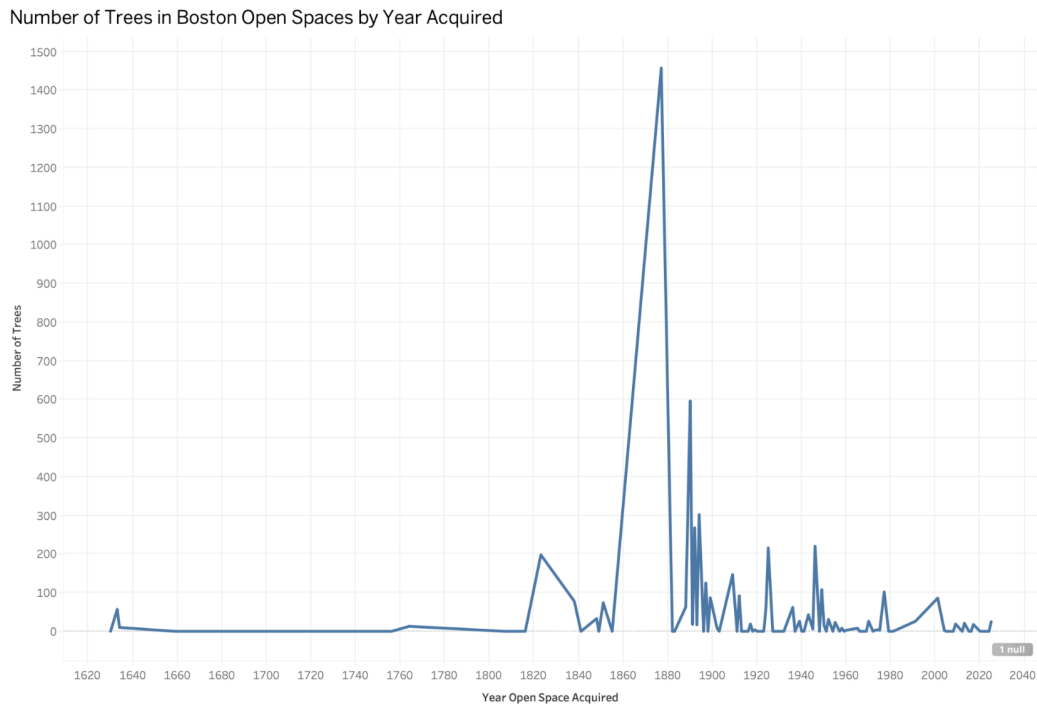
The following is the task table provided for my analysis:

Task #	Task	High-level	Mid-level	Low-level
1	What do these data of trees planted in Boston look like over time (i.e., how has the planting	Analyze → Consume → Present	Search → Explore	Query → Compare

	evolved or progressed over time)?			
2	Where are the trees planted? (Which neighborhoods/parks?)	Analyze → Consume → Present	Search → Locate	Query → Identify
3	What types of trees have been planted?	Analyze → Consume → Present	Search → Lookup	Query → Summarize
4	What is the benefit to the Boston ecosystem from the planted trees comparing today versus a projected model of 20 years in the future in terms of: 1. Increase in canopy cover 2. Mitigation of storm water run-off and flooding 3. Air quality improvement (reduction of CO2)	Analyze → Consume → Present	Search → Explore	Query → Compare

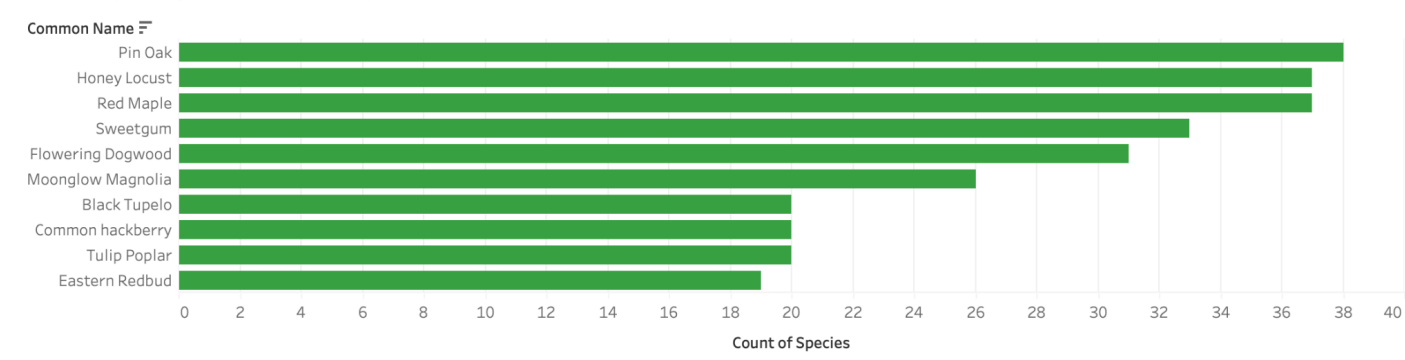
In my report, I decided to focus primarily on Task 2 and 3, which involved analyzing the geographic distribution and species composition of trees in Boston. These tasks were the most supported by the available data. Task 1 was partially addressed by replacing the "date_plant" column with "YearAcquired" as the temporal column, given the limited data points in the former. Overall, Task 2 and 3 together provide the clearest picture of Tree Boston's past planting activities and recommendations for the future, which I will discuss below.

Observations & Insights



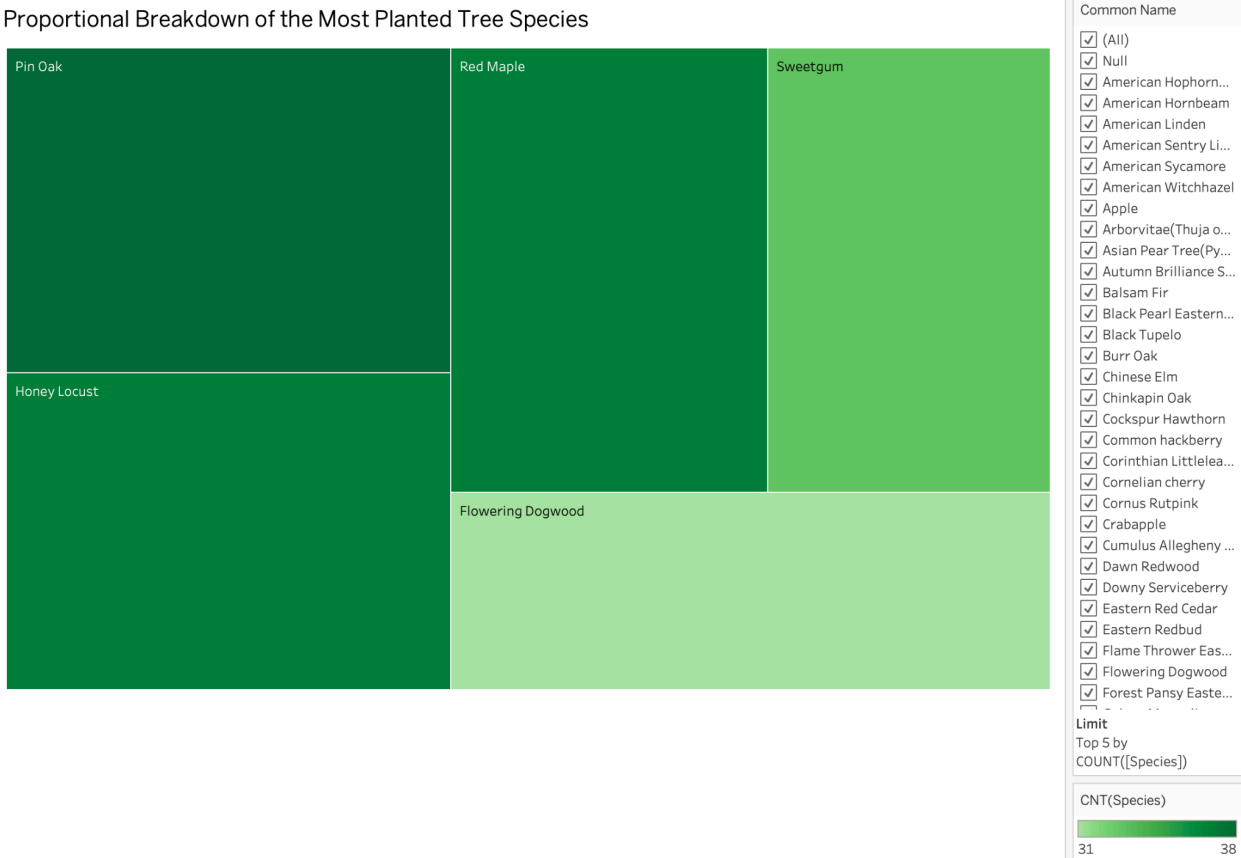
My initial visualization was a line chart depicting the number of open spaces acquired in Boston over the past couple of centuries. As evident in my visualization, there was a significant spike, which occurred because Boston hired Frederick Law Olmsted to design and develop the Emerald Necklace in Boston after his role in designing Central Park. However, as you can observe, Tree Boston has been actively helping to restore that number after a substantial decline following the year 2000.

Most Frequently Planted Tree Species in Boston



For my second visualization, I opted to use a bar chart to depict the 10 most frequently planted trees across Boston. Red Maple stands out as the most popular choice, followed closely by Pin Oak and Tulip Poplar. This helps simplify all 152 rows of data in a more digestible way allowing the user to spot patterns more clearly.

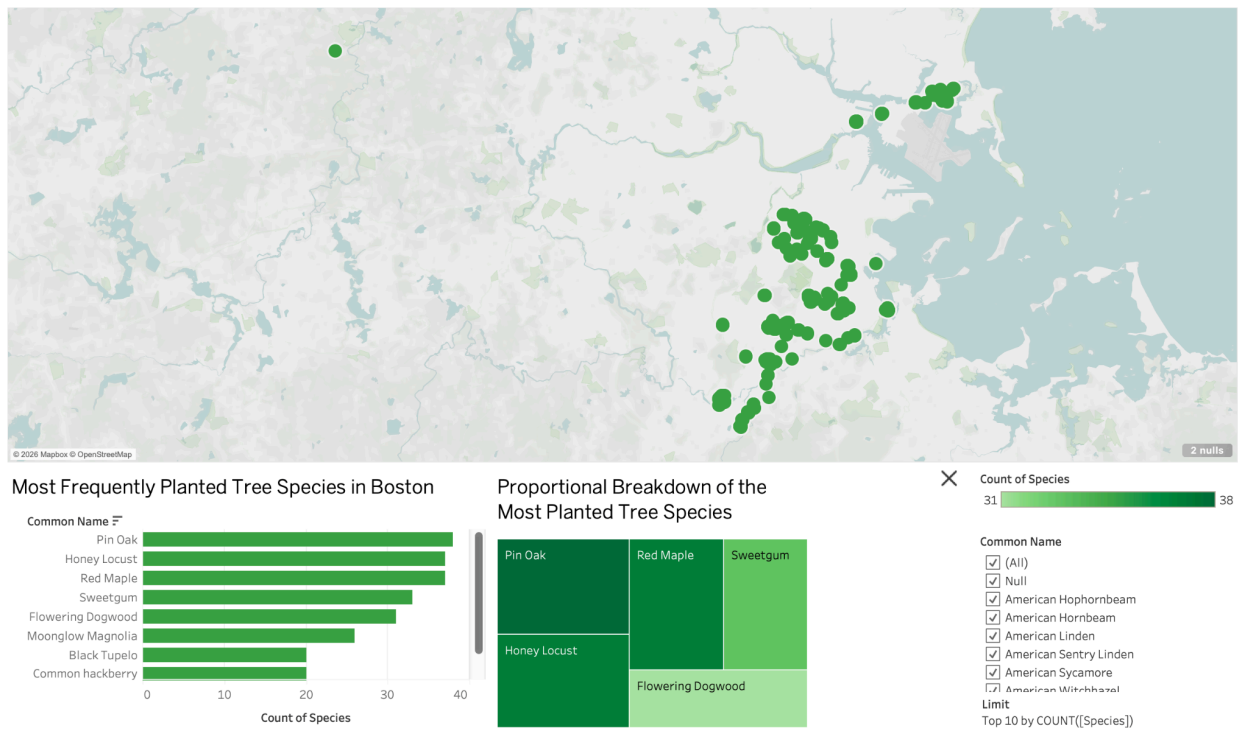
Proportional Breakdown of the Most Planted Tree Species



My third visualization encodes species counts using size and rank, with color intensity representing the top species planted in Boston. To enhance readability in the high-level dashboard below, I've chosen to focus on five species. Additionally, I've incorporated an interactive species selector, allowing users to filter by their preferred trees. As you can see, Red Maple, Pin Oak, and Flowering Dogwood are the most popular trees in Boston's tree canopies.

TreeBoston Planting Dashboard with Species and Location Explorer

Geographic Distribution of TreeBoston Plantings Across Boston



The dashboard seamlessly integrates three views map, bar chart, and treemap, into a single interactive interface, allowing users to explore and manipulate the data. Additionally, I incorporated the brushing and linking techniques we demonstrated in class, enabling users to select any species in any row and visualize the data in a more

organized manner. A species dropdown filter on the right panel provides users with the option to filter the data based on their interest in specific species. I also opted to use different shades of green as the primary color to ensure color-blind users can still comprehend the visualizations. The dashboard effectively highlights that the most planted species are not evenly distributed across the city. By filtering the data by species, users can observe cluster patterns and gain deeper insights into the distribution of plant species.

Conclusion

The report of Tree Boston Planting Data shows that most of the trees in their canopy are either in the south end, Mattapan, or Dorchester, which aligns with Tree Boston's environmental justice mission to serve minority majority neighborhoods. These urban areas are more exposed to heat, so it's very important that Tree Boston is doing its part to provide the community with its needs. According to my analysis, the most dominant tree species planted is red maple, followed closely by pin oak. And lastly, although we had data quality issues, I tried to make temporal analysis possible with the data.

Again, I really enjoyed working with Tree Boston and volunteering my time to build a dashboard that will hopefully be adopted and used in the future. If I had any suggestions, it would be maybe creating a process that would track survival rates of the trees to measure the long-term impact and not just the growth process. I also mentioned this above but also standardizing the date_plant column to be the correct longitude and latitude to plant the trees as well as fact-checking so I don't get messy results like in my map visualization in Tableau. Overall, I loved working with Tree Boston and can't wait to see what they have in store for the future.
